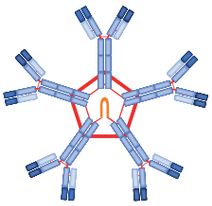
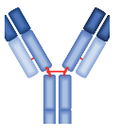
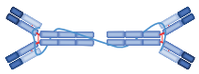
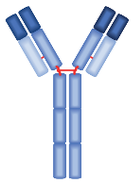
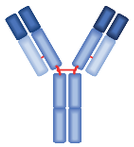
10.5.2. Types of Antibodies - Classification

The Five Immunoglobulin(Ig) Classes					
	IgM Pentamer	IgG Monomer	Secretory IgA Dimer	IgE Monomer	IgD Monomer
			 Secretory Component		
Heavy chains	μ	γ	α	ε	δ
Number of antigen binding sites	10	2	4	2	2
Molecular weight (Daltons)	900000	150000	385000	200000	180000
Percentage of total antibody in serum	6%	80%	13%	0.002%	1%
Crosses placenta	No	Yes	No	No	No
Fixes complement	Yes	Yes	No	No	No
Fc binds to		Phagocytes		Mast cells land basophils	
Function	Main antibody of primary responses, best at fixing complement, the monomer form of IgM serves as the B cell receptor.	Main blood antibody of secondary responses, neutralizes toxins, opsonisation	Secreted into mucus, tears, Saliva, Colostrum	Antibody of allergy and antiparasitic activity	B Cell receptor

10.6 ANTIGEN-ANTIBODY REACTIONS

Antigen-Antibody reactions refer to the interaction between antigens and antibodies. The reactions between antigens and antibodies occur in three stages.

The primary stage is the initial interaction between antigen and antibody without any visible effects.

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